Rationale:

During my sophomore year, I volunteered at a cancer awareness fundraiser, where the researchers discussed the various treatment pathways. That is where I learned that the oncologist deciding when to begin the treatment, in part, is making a mathematical prediction about how fast a tumour will grow. I had always thought of cancer as a purely biological problem, so the idea that a differential equation could help with a clinical decision surprised me.

During this fundraiser, the professor explained that, specifically, Lewis Lung Carcinoma (LLC) allows researchers to study how cancers behave and respond to treatments. LLC acts as a “guinea pig” to help researchers measure tumor growth over time and collect reproducible experimental data in mice. Although LLC originates from mice, it shares the same characteristics as aggressive human cancers, helping scientists study tumor progression and spread without having to monitor human cancer for multiple years.

Understanding tumor growth is important because doctors and researchers can then estimate how aggressive a cancer is, how quickly it may spread, and when intervention is necessary. However, tumors don’t keep growing forever; Carrying Capacity determines the tumor’s limitations inside the body. At first, the tumor is small and resources are abundant, so it grows quickly. But as it becomes larger, the tumor starts competing with itself for resources.

Literature describes the Gompertz model and logistic model to explain tumor growth. Both models highlight that growth slows over time, but they slow down in different mathematical ways. Solving each model for the carrying capacity determines which model produces the most accurate description of the LLC data. This investigation explores how accurately each model describes real measurements, and which is more biologically realistic.

Introduction:

Tumor growth can appear exponential because a larger tumor contains more cells. This means that the tumor contains more cells that can divide and cause more growth. However, tumor growth isn’t exponential because as they increase in size, growth slows down due to limited resources such as oxygen availability, immune response, angiogenesis, and other biological constraints. This is why we can model tumor growth with differential equations that create a sharp increase in volume, followed by a slow decrease. This investigation will compare the logistic and Gompertz growth models. Both models describe tumor volume, V, in mm^3, as a function of time t in days.

Variables and Data

The table below holds the variables for this investigation. These variables were taken from the Vaghi et al experiment in 2020.

Symbol	**Definition**	**Units**	**Literature value (Initial Value)**
V	Tumor Volume	mm^3	measured
t	Time since injection	days	measured
V_0	Initial tumor volume	mm^3	1.0
r (logistic)	Logistic growth rate constant	day^{-1}	0.5
r (Gompertz)	Gompertz growth rate constant	day^{-1}	0.10
K	Carrying capacity	mm^3	Initially unknown, must find through this investigation.

The growth rate constants above are the published starting values; the regression later refines them. Furthermore, both models use the letter r, but the value is not the same in each, because the two equations slow growth in different ways. In the logistic model, r sets the early exponential rate before the slowing factor takes hold; in the Gompertz model, r controls how fast the logarithmic slowing term decays. Each value comes from a separate published fit to LLC.

The Vaghi et al. 2020 dataset has eleven tumor volume measurements from one LLC Tumor. It is recorded between days 5 and 20 after injection

t days	V (mm^3)
5	54.9
6	59.4
7	85.7
11	261.6
12	412.3
13	488.5
14	618.9
15	799.0
18	1218.7
19	1415.5
20	1410.1

Because the dataset records tumor volumes to one decimal place, I report carrying capacities to the nearest whole mm^3, growth-rate constants to three significant figures, and R^2 to four decimal places so the two competing fits can be distinguished.

For this investigation, we must use the data points to estimate the value of K, the carrying capacity. K represents the limiting tumor volume predicted by each model: the logistic versus the Gompertz. Graphically, K is the horizontal asymptote of each model, which can help identify which model is the most reasonable for modeling growth of Lewis Lung Carcinoma. The value of K must be significantly above the largest observed tumor volume because the mice in the Vaghi et. al study are euthanized before the full development of the tumor due to ethical reasons.

This leads me to my research question: How accurately do the logistic and Gompertz differential equation models describe the growth of Lewis Lung Carcinoma tumors, and which model provides a more biologically realistic representation?

To acquire results, I will start by deriving the Logistic and Gompertz equations from their differential equation, using the literature growth rate constants as initial estimates and fixing the initial volume. I will first estimate the carrying capacity from individual data points, then refine both the growth rate constant and the carrying capacity through nonlinear Desmos regression to find the best-fitting equation for each model. To conclude the investigation, I will compare the Gompertz model versus the Logistic model to see which model meets the biological standards and the current knowledge of cancer growth.

<u>The Logistic Growth Model:</u>

The logistic model creates an S-shaped curve called a sigmoid curve, and the graph is symmetric around the inflection point, meaning the curve before the inflection point mirrors the curve after it.

The logistic model is written as the differential equation:

$$\frac{dV}{dt} = rV(1 - \frac{V}{K})$$

In this equation, $\frac{dV}{dt}$, is the rate of change of tumor volume with respect to time, V is the tumor volume, t is time in days, r is the logistic growth-rate constant, and K is the carrying capacity.

The model assumes that when V is less than K, the tumor grows almost exponentially. As V approaches K, the factor $1 - \frac{V}{K}$ approaches 0, causing growth to slow. To make this work for K, we have to solve the differential equation.

Starting Equation:

$$\frac{dV}{dt} = rV(1 - \frac{V}{K})$$

Separate the variables:

$$\frac{dV}{V(1-\frac{V}{K})} = r_L dt$$

Rewrite the denominator:

$$V(1 - \frac{V}{K}) = \frac{V\,(K - V)}{K}$$

So:

$$\frac{K}{V(K-V)}(dV) = r(dt)$$

Using partial fractions:

$$\frac{K}{V(K-V)} = \frac{1}{V} + \frac{1}{K - V}$$

Now we get:

$$\int(\frac{1}{V} + \frac{1}{K - V})dV = \int rdt$$

$$ln|V| - ln|K - V| = rt + C$$

Using logarithmic rules:

$$ln(\frac{V}{K - V}) = rt + C$$

Raising both sides as powers of e, and using A as a constant for replacement of C:

$$\frac{V}{K - V} = Ae^{rt}$$

Solving for V as a function of time, t:

$$V = Ae^{rt}(K - V)$$

$$V = AKe^{rt} - AVe^{rt}$$

$$V(1 + Ae^{rt}) = AKe^{rt}$$

$$V(t) = \frac{AKe^{rt}}{1 + Ae^{rt}}$$

Dividing numerator and denominator by Ae^{rt}, the logistic solution is:

$$V(t) = \frac{K}{1 + Be^{-rt}}$$

Where B is a constant determined by the initial condition. In this solution, B = $\frac{1}{A}$.
Here V_0 = 1 mm^3 is the conventional initial volume at injection (t = 0); the first measurement, 54.9 mm^3 on day 5, occurs after several days of growth, so it sets the model's starting size rather than replacing the initial condition.

Applying V(0) = 1:

$$1 = \frac{K}{1+B}$$

$$1 + B = K$$

$$B = K - 1$$

$$V(t) = \frac{K}{1 + (K-1)e^{-rt}}$$

Since r = 0.5:

$$V(t) = \frac{K}{1 + (K-1)e^{-0.5t}}$$

<u>Solving for K from a data Point:</u>
Starting with:

$$V(t) = \frac{K}{1 + (K-1)e^{-0.5t}}$$

To estimate the carrying capacity, I rearranged the equation in terms of K. First, I multiplied both sides by the denominator:

$$V(1 + (K - 1)e^{-r_L t}) = K$$

Expanding the equation:

$$V + VKe^{-r_L t} - Ve^{-r_L t} = K$$

Moving all terms of K to one side:

$$V - Ve^{-r_L t} = K - VKe^{-r_L t}$$

Factor:

$$V(1 - e^{-r_L t}) = K(1 - Ve^{-r_L t})$$

Therefore, by dividing:

$$K = \frac{V(1 - e^{-r_L t})}{1 - Ve^{-r_L t}}$$

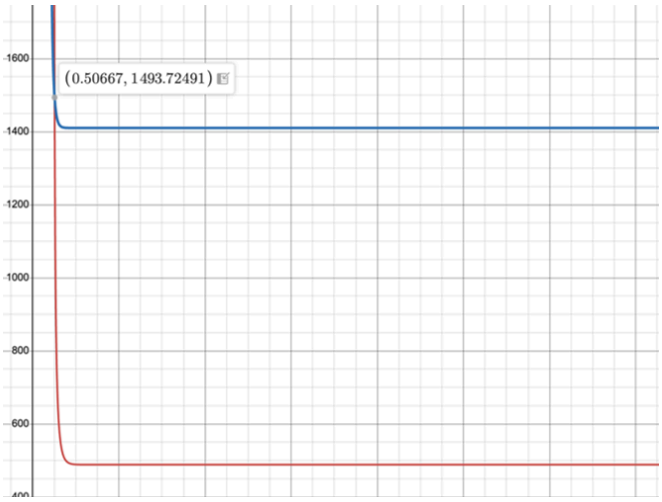
This equation can be used to estimate K from any data point where $1 - Ve^{-rt} > 0$. Since we have an equation for K(r), we can use systems of equations no mmm to solve for both values. Using the data points below, we can create a system and graph using Desmos to find K and r.

t days	V (mm^3)
13	488.5
20	1410.1

$$K = \frac{488.5\,(1 - e^{-r(13)})}{1 - 488.5e^{-r(13)}} \text{ (red line)}$$

$$K = \frac{1410.1\,(1 - e^{-r\,(20)})}{1 - 1410.1e^{-r(20)}} \text{ (blue line)}$$

Each data point produces a different curve of K as a function of r. The growth rate that makes both curves agree is the single r consistent with both observations, so the intersection of the two curves gives the (r, K) pair that satisfies the model at both points simultaneously.

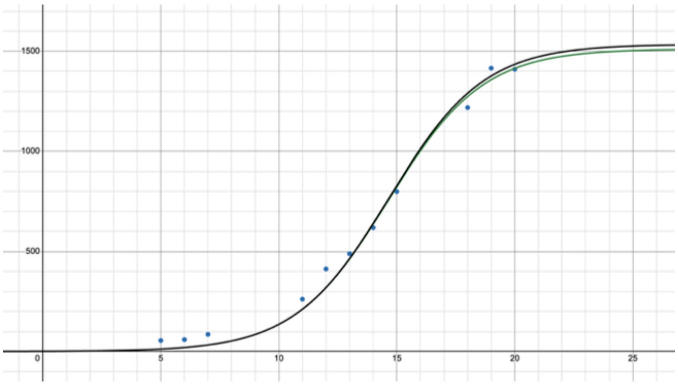


With K equaling 1494 mm^3 and r equaling 0.507, we have the calculated logistic model of LLC as:

$$V(t) = \frac{1494}{1 + 1493e^{-0.507t}}$$

Even though the equation is solved, using two of the data points is not the best-fitting logistic equation. To attain a more accurate line, the use of nonlinear regression is needed. With eleven data points, the use of Desmos graphing technology is needed to algorithmically regress the points along the raw logistic equation.

This is done using the method of least squares, which takes the vertical distance between a predicted line and a point and squares it, creating the line where the sum of these squares is at the absolute minimum.



Key:
Green = Desmos-regression equation
Black = Averaged K value equation

Desmos graphing technology gave the equation:

$$V(t) = \frac{1496}{1 + 1495e^{-0.504t}}$$

This investigation will use the Desmos regression model with a K-value of 1496 mm^3, as it had a higher coefficient of determination of 0.9910, compared to 0.9908 from solving the system of equations.

The Gompertz Growth Model

The Gompertz Growth Model is given by the equation:

$$\frac{dV}{dt} = r_G V ln(\frac{K}{V})$$

The Gompertz model slows growth using the logarithmic term $\ln(\frac{K}{V})$. When V is smaller than K, the logarithmic term is large, allowing for faster growth. As V approaches K, $\ln(\frac{K}{V})$ approaches 0, slowing growth. The key difference between logistic and Gompertz is that the logistic equation is symmetrical about the inflection point, while Gompertz is not.

To solve this differential equation, we must start with a u-substitution:

$$Letting\ u = ln(\frac{K}{V}) = ln(K) - ln(V)$$

Differentiate with respect to t. Since K is a constant, $\frac{d}{dt}(ln(K)) = 0$

$$\frac{du}{dt} = -\frac{1}{V}(\frac{dV}{dt})$$

Substitute the differential equation:

$$\frac{du}{dt} = -\frac{1}{V}(r_G V ln(\frac{K}{V}))$$

$$\frac{du}{dt} = -r_G u$$

$$\frac{du}{u} = -rdt$$

Integrate:

$$ln(u) = -r_G t + C$$

Raise both sides as powers of e:

$$u = Be^{-r_G t}$$

Substitute u back:

$$ln(\frac{K}{V}) = Be^{-r_G t}$$

Raising both sides as powers of e to get it in terms of V:

$$\frac{K}{V} = e^{Be^{-r_G t}}$$

$$V = Ke^{-Be^{-r_G t}}$$

Therefore:

$$V(t) = Kexp(-\ Be^{-r_G t})$$

Using the initial condition $V(0) = V_0 = 1$:

$$1 = Kexp(-\ B)$$

$$\frac{1}{K} = e^{-B}$$

Taking the natural log:

$$-\ ln(K) = -\ B$$

$$B = ln(K)$$

So the Gompertz model becomes:

$$V(t) = Kexp[-\ ln(K)e^{-rt}]$$

Since r = 0.10,

$$V(t) = Kexp[-\ ln(K)e^{-0.10t}]$$

To solve for K, first take the natural log of both sides:

$$ln(V) = ln(K) - ln(K)e^{-rt}$$

Factor ln(K):

$$ln(V) = ln(K)(1 - e^{-r_G t})$$

Divide to get it in terms of ln(K):

$$ln(K) = \frac{ln(V)}{1 - e^{-rt}}$$

Raise both sides as powers of e:

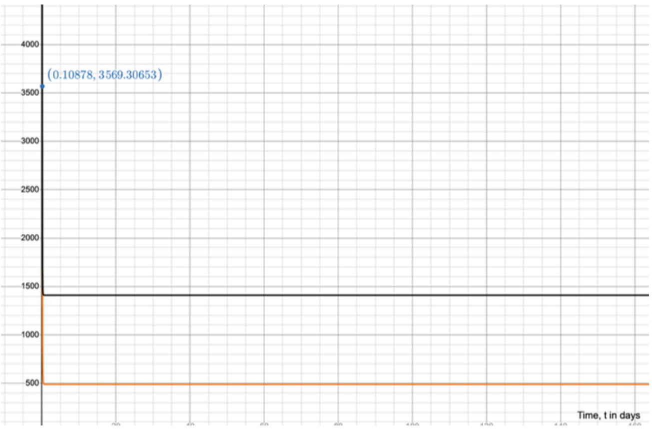
$$K = exp(\frac{ln(V)}{1 - e^{-r_G t}})$$

With the equation now in K(r_G), we can solve for K and r_G using a system of equations.

t days	V (mm^3)
13	488.5
20	1410.1

$$K \ = \ exp(\frac{ln(488.5)}{1-e^{-r_G(13)}})$$ - Yellow line on Graph

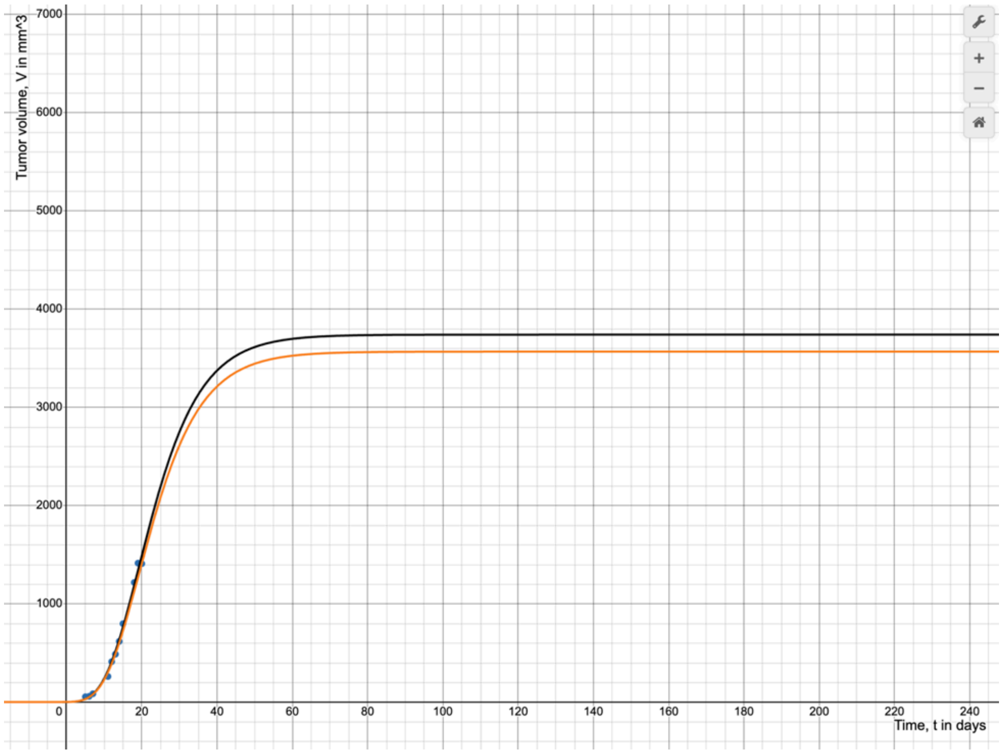
$$K \ = \ exp(\frac{ln(1410.1)}{1-e^{-r_G(20)}})$$ - Black Line on Graph



With K equaling 3569 mm^3 and r_G equaling 0.109, the calculated Gompertz model of LLC is:

$$V(t) \ = \ 3569e^{-8.18e^{-0.109t}}$$

Using Desmos-based graphing technology, we can perform regression (Black line) and compare it to the calculated Gompertz model (yellow line), and we see that they are very similar. As in the logistic case, the intersection of the two curves gives (r_G, K) pair consistent with both data points.



The technology gave the equation:

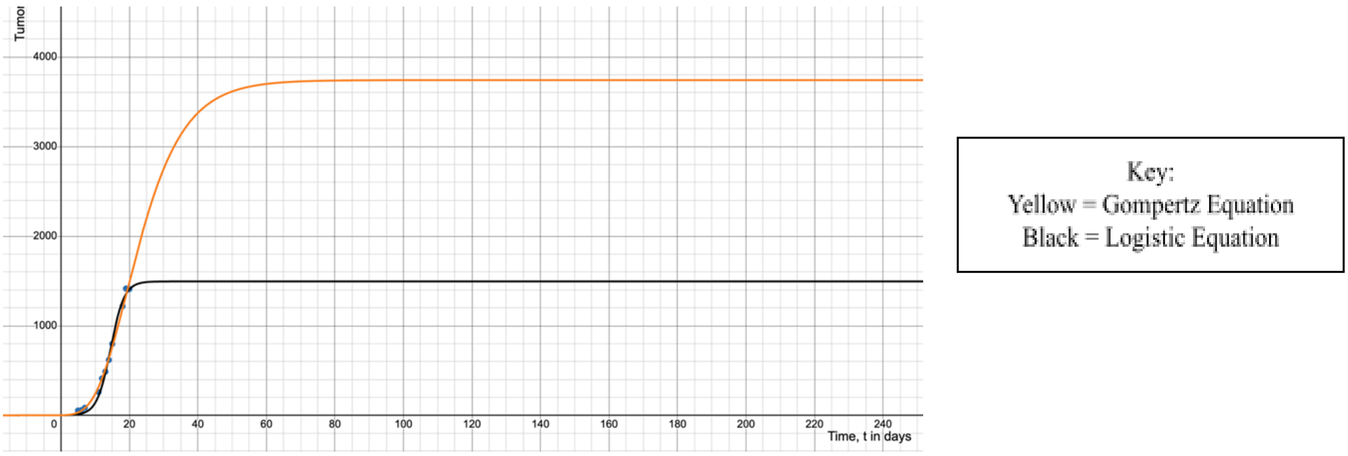
$$V(t) \ = \ 3742e^{-8.23e^{-0.11t}}$$

For this investigation, we will continue with the Desmos-regressed model with a K-value of 3742 mm^{3}, as the R^2 fit value of it is better than our equation: 0.9931 and 0.9866, respectively.

Using the two completed equations:

$$V_{Logistic}(t) = \frac{1496}{1 + 1495e^{-0.504t}}$$

$$V_{Gompertz}(t) = 3742e^{-8.23e^{-0.11t}}$$



Graph comparing Logistic and Gompertz tumor growth models

The investigation shows that the logistic model has a carrying capacity of 1496 mm3, while the Gompertz model has a carrying capacity of 3742 mm^3. Although both values exceed the largest observed tumor volume of 1415.5 mm^3, the Gompertz model has a significantly larger gap between the largest observed tumor and the carrying capacity. Keeping in mind that the euthanization of the mice causes the premature deaths of the tumor, the logistic model doesn't give the tumor more time and space to continue growing after the final data point. On the other hand, the Gompertz model predicts a much higher carrying capacity that is biologically plausible and indicative of the limit the tumor would approach.

Through the graph, it may seem as though the logistic model fits better to the data, but it doesn't follow standard tumor growth patterns, as it predicts that growth will flatten midway through the tumor's life span.

To mathematically prove this point, we can compare the inflection points of the two models, finding the volume at which each tumor is growing the fastest. The inflection point occurs where the growth rate $\frac{dV}{dt}$ is greatest, so for each model, I differentiate $\frac{dV}{dt}$ with respect to V and set it to zero.

Logistic

The growth rate as a function of V:

$$\frac{dV}{dt} = rV\left(1 - \frac{V}{K}\right) = rV - \frac{rV^2}{K}$$

Differentiate with respect to V and set equal to zero:

$$\frac{d}{dV}\left(\frac{dV}{dt}\right) = r - \frac{2rV}{K} = r\left(1 - \frac{2V}{K}\right) = 0$$

$$\Rightarrow \quad V = \frac{K}{2}$$

Gompertz

The growth rate as a function of V:

$$\frac{dV}{dt} = r_G\, V \ln\left(\frac{K}{V}\right)$$

Differentiate (product rule, with $\frac{d}{dV}\ln\frac{K}{V} = -\frac{1}{V}$) and set to zero:

$$\frac{d}{dV}\left(\frac{dV}{dt}\right) = r_G\left[\ln\frac{K}{V} + V\left(-\frac{1}{V}\right)\right] = r_G\left[\ln\frac{K}{V} - 1\right] = 0$$

$$\Rightarrow \quad \ln\frac{K}{V} = 1 \quad \Rightarrow \quad V = \frac{K}{e}$$

With $K = 1496$: $V = 748\ \mathrm{mm}^3$ *(logistic).* *With* $K = 3742$: $V \approx 1376\ \mathrm{mm}^3$ *(Gompertz).*

For the logistic model, this gives $V = \frac{K}{2}$. With K = 1496 mm^3, the inflection is at V = 748 mm^3. The LLC data passes this volume between days 14 and 15, yet the logistic model claims the tumor should already be decelerating there. This doesn't fit, as the mouse was euthanized while the tumor was still growing fast. The Gompertz model shows the inflection is at $V = \frac{K}{e}$. With K = 3742 mm^3, V is about 1376 mm^3, close to the largest observed tumor (1415.5 mm^3). The tumor was therefore still near its fastest-growing phase when the experiment ended.

<u>Reflection:</u>

The clearest idea from this exploration is that a model can fit the data you happen to have and still describe the world poorly. The graph created through the data points favors the logistic model, but its carrying capacity was biologically impossible, so visually, this would've led me to the wrong conclusion.

However, there are limitations to what I did. I used the literature growth rate constants only as starting values and then let the regression refine both r and K, so my fitted r values (0.504 for logistic, 0.11 for Gompertz) differ slightly from the published ones. Because the logistic carrying capacity is far more sensitive to r than the Gompertz value is, even small differences in the fitted r propagate more strongly into the logistic K.

I also only analyzed a single mouse, which in turn means one tumor. Therefore, I can't say the result holds for every LLC tumor. It is also worth mentioning that neither model captures the real biology; blood supply, immune system, cell death, and nutrient limits are all compressed into a single equation. Many underlying factors weren't addressed in this investigation. If the mouse wasn't eating properly on some days versus others, the tumor growth may have changed.

For further investigation, I could potentially use numerical regression to fit both r and K together and compare the models with a goodness-of-fit measure such as RMSE, and I would repeat the whole process across many tumors to see whether Gompertz is consistently the better description. However, Gompertz is stipulated to be the better model in terms of tumor growth.

These limitations may have caused minor variances in our data, but overall the investigation reached its goal. The carrying capacity proved that the Gompertz model gives a more biologically accurate estimate when compared to the Logistic model. Gompertz allows earlier growth acceleration while maintaining a larger final carrying capacity, proving what literature knows so far about cancer.

Conclusion:

The goal of this investigation was to see which model better describes Lewis Lung Carcinoma (LLC) growth when the initial tumor volume is fixed and the growth rate constants begin from literature values but are refined by regression.

The logistic model produced: $K_{Logistic} \approx 1496\ mm^3$

The Gompertz Model produced: $K_{Gompertz} \approx 3742\ mm^3$

The two models produce opposite trade-offs: the logistic model fits the observed data more closely but fails biologically, while the Gompertz model is marginally looser in fit but produces a biologically credible carrying capacity. In terms of accuracy, the logistic model fits the measured data slightly better; in terms of biological realism, it fails badly, predicting a carrying capacity of 1496 mm^3 that sits right on top of a tumor that was still growing. The Gompertz model is marginally less tight to the data but predicts a carrying capacity of 3742 mm^3, comfortably above the measurements and consistent with a tumor whose growth was cut short by euthanization, and it produces valid estimates from all eleven data points.

The Gompertz model's inflection point at $V = \frac{K}{e} \approx 1376\, mm^3$, near the final observed volume, confirms that the tumor was still in its rapid-growth phase at euthanization, consistent with ethical experimental constraints. The logistic model's inflection at $V = \frac{K}{2} \approx 748\, mm^3$, crossed by day 14 to15, implies the tumor had already begun decelerating, which contradicts the observed data trajectory. This makes the Gompertz model the more biologically defensible choice.

Works Cited

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